AGOP: A Computable Decision Procedure for Multi-Criteria Selection Under Uncertainty

Abstract

We specify, test, correct, and prove a deterministic decision procedure—AGOP (Attributed-Graph Operator Pipeline)—for selecting among a finite set of alternatives under multi-criteria, strategic, structural, and stochastic considerations. AGOP composes graph-theoretic, game-theoretic, systems-theoretic, topological, deductive, stochastic, and probabilistic operators in a typed directed acyclic graph with two nested fixed-point subroutines. The procedure is domain-agnostic: it applies to any decision problem in which (i) alternatives can be attributed along multiple axes, (ii) alternatives interact through a compatibility relation, (iii) a decision-maker and at least one counter-party have potentially divergent interests, and (iv) outcomes are uncertain. We exhibit one arithmetic evaluation on a thirteen-alternative reference instance, demonstrate that the unmodified draft algorithm fails three specification tests, repair the failures, and prove determinism, well-typedness, convergence, and output admissibility for the corrected algorithm. Complexity is polynomial in all parameters except the Nash subroutines, which are constant-time for the two-player case and linear in the number of surviving alternatives for the adoption case.

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1. Problem statement

1.1 Input

A decision problem $P = (V_0, \mathcal{A}, \Theta, \mathcal{R})$ where:

- $V_0 = \{v_1, \ldots, v_n\}$ is a finite set of decision alternatives;
- $\mathcal{A} = \{x_1, \ldots, x_d\}$ is a finite set of decision axes, each a real-valued function on V_0;
- Θ is a parameter vector containing utility parameters, admissibility thresholds, prior distributions for stochastic evaluation, and normalization constants;
- $\mathcal{R} \subseteq V_0 \times V_0$ is a symmetric compatibility relation: $(u, v) \in \mathcal{R}$ iff alternatives u and v share at least one relevant attribute or resource.

The decision axes are not assumed independent. The compatibility relation is not assumed transitive. The parameter vector is not assumed known with certainty; it is specified and its consequences are propagated through the procedure.

1.2 Output

A ranking $R = (v_{(1)}, \ldots, v_{(k)})$ of alternatives $v_{(i)} \in V_0$, each annotated with:

- a structural equivalence class $\tau(v_{(i)}) \in \mathcal{T}$, where $\mathcal{T}$ is the set of classes induced by the procedure;
- an outcome estimate $\hat{Y}(v_{(i)})$ with a lower confidence bound $\mathbf{LCB}_{1-\alpha}(\hat{Y}(v_{(i)}))$;
- an adoption status $\nu^*(v_{(i)}) \in [0, 1]$.

The output is admissible if and only if every $v_{(i)} \in R$ satisfies: (i) $v_{(i)}$ survives three deductive filters; (ii) $v_{(i)}$ supports an equilibrium in the two-player game between decision-maker and counter-party; (iii) $v_{(i)}$ has $\mathbf{LCB}_{1-\alpha}(\hat{Y}(v_{(i)})) \geq 0$; (iv) $v_{(i)}$'s adoption is supported by a population equilibrium over the survivor set.

1.3 Objective

Maximize decision value $Y = \mathbf{reward} - \mathbf{cost}$ subject to the admissibility conditions of §1.2. The procedure does not learn parameters; it evaluates consequences of given parameters.

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2. Formal objects

2.1 Attributed directed graphs

Let $G = (V, E, \mathbf{X})$ be an attributed directed graph where $V \subseteq V_0$, $E = \{(u, v) : (u, v) \in \mathcal{R}|_V\}$, and $\mathbf{X} \in \mathbb{R}^{n \times d}$ with row i equal to $x(v_i) = (x_1(v_i), \ldots, x_d(v_i))$.

Each column is L^2-normalized without loss of generality: $\|x_j\|_2 = 1$ for all j.

2.2 Orthogonal axes

Define $\mathbf{C} = \mathbf{Corr}(\mathbf{X}) \in \mathbb{R}^{d \times d}$. The axes are δ-orthogonal if $\|\mathbf{C} - \mathbf{diag}(\mathbf{C})\|_F < \delta$. If the condition fails, replace $\mathbf{X}$ by the SVD-rotated form $\mathbf{X}' = U\Sigma$ where $\mathbf{X} = U\Sigma V^\top$ is the thin SVD.

Remark 2.1 (interpretation). The rotated columns are principal components, not interpretable decision axes. The procedure applies orthogonalization only to the submatrix $\mathbf{X}_{D^*}$ of surviving alternatives, not to the full $\mathbf{X}$, preserving axis labels for the substantive steps.

2.3 State

A state is a tuple $s_t = (G_t, \gamma_t, \Pi_t, \mathcal{E}_t, \Sigma_t, \mathcal{T}_t, D_t, \nu_t^*, \hat{P}_t, A_t)$ for $t \in \{0, \ldots, 11\}$, with s_0 the empty state and s_{11} the terminal state. Each field is computed by the phase indexed t.

2.4 Conventions (determinism)

C1 (tie-breaking). All $\mathbf{arg\,max}$ operations resolve ties lexicographically by alternative identifier.

C2 (equilibrium selection). When $\mathcal{E} \neq \emptyset$ has multiple elements, select $e^* \in \mathbf{arg\,max}_{e \in \mathcal{E}}[u_{\mathrm{DM}}(e) + u_{\mathrm{CP}}(e)]$. If multiple elements attain the maximum, select lexicographically. This is the utilitarian selection under convention C1.

C3 (stochastic seed). All M draws in the stochastic phase use a fixed seed specified in Θ.

C4 (resonance target). The resonance Gaussian is *targeted*, not mean-centred: its centre is $(r^*, b^*) = (\mathbf{max}_v\, r(v), \mathbf{min}_v\, b(v))$ after normalization to $[0, 1]$, with $r = 1$ denoting full readiness and $b = 0$ denoting no barrier.

C5 (ranking score). The ranking orders by $\Phi(v) = A(v) \cdot \mathbf{LCB}_{1-\alpha}(\hat{Y}(v))$, not by $A(v)$ alone.

C6 (adoption game). The adoption game has binary actions $a_v \in \{0, 1\}$ with capacity constraint K.

3. The AGOP algorithm

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Input: P = (V_0, A, θ, R); conventions C1–C6.
Output: R = (v_(1), …, v_(k)) with annotations.

Phase 0.   (G_0, X) ← S(P)                        # attribute the alternatives
Phase 1.   γ       ← C(G_0)                       # structural centrality
Phase 2.   Π       ← Γ(G_0, γ; θ)                 # payoff construction
Phase 3.   e_1     ← N_1(Π; C2)                   # strategic equilibrium
Phase 4.   (Σ, β)  ← Σ_op(G_0, γ)                 # systems annotation
Phase 5.   T       ← Θ_op(G_0, X)                 # topological classes
Phase 6.   D*      ← Δ(G_0, θ; C1)                # deductive filter
Phase 7.   v*      ← N_2(D*, Π; C6, θ)            # adoption equilibrium
Phase 8.   P_hat   ← M(D*, v*; C3, θ)             # stochastic evaluation
Phase 9.   A       ← H(D*, X, P_hat; C4)          # target proximity
Phase 10.  Φ       ← A(v) · LCB_{1-α}(P_hat(v))   # composite score
Phase 11.  R       ← TopK(Φ, k; C1)               # output
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4. Operator specifications

4.1 Search $\mathcal{S}$

$$\mathcal{S} : P \mapsto (G_0, \mathbf{X}) \quad \text{where} \quad x_j(v) = f_j(\text{source}_j, v),$$

with f_j a documented measurement map for axis j. Each data point passes a triple validity check (traceable, consistent with at least one independent source, recent within horizon T). Points failing any check are set to the neutral value $\tilde{x}_j = \mathbb{E}[x_j]$ and flagged.

4.2 Centrality $\mathcal{C}$

$$\gamma(v) = \alpha_1 \frac{d(v)}{n-1} + \alpha_2 B(v) + \alpha_3 E(v), \qquad \alpha_1 + \alpha_2 + \alpha_3 = 1, \quad \alpha_i \geq 0,$$

where $d(v)$ is out-degree in G_0, $B(v)$ is normalized betweenness (divided by $(n-1)(n-2)$), and $E(v)$ is eigenvector centrality normalized so $\max_v E(v) = 1$. All three terms lie in $[0,1]$, so $\gamma(v) \in [0,1]$.

4.3 Payoff Γ

Two-player, finite, non-cooperative:

$$\mathcal{A}_{\text{DM}} = \{\text{bespoke}, \text{standard}\}, \quad \mathcal{A}_{\text{CP}} = \{\text{adopt}, \text{demand}\}.$$

$$\Pi = \begin{pmatrix} (u^{bb}_{\text{DM}}, u^{bb}_{\text{CP}}) & (u^{bd}_{\text{DM}}, u^{bd}_{\text{CP}}) \\ (u^{sb}_{\text{DM}}, u^{sb}_{\text{CP}}) & (u^{sd}_{\text{DM}}, u^{sd}_{\text{CP}}) \end{pmatrix}$$

parametrized by $\theta = (c_b, c_s, w, \ell)$ where $c_b > c_s$ (bespoke vs. standard decision-maker cost), w is counter-party willingness-to-pay, and ℓ is liability exposure under autonomous operation:

$$\begin{aligned} u^{bb}_{\mathrm{DM}} &= w - c_b, & u^{bb}_{\mathrm{CP}} &= \bar{v} - w, \\ u^{bd}_{\mathrm{DM}} &= -c_b, & u^{bd}_{\mathrm{CP}} &= 0, \\ u^{sb}_{\mathrm{DM}} &= w - c_s, & u^{sb}_{\mathrm{CP}} &= \bar{v} - w - \ell, \\ u^{sd}_{\mathrm{DM}} &= -c_s, & u^{sd}_{\mathrm{CP}} &= 0. \end{aligned}$$

4.4 Market Nash $\mathcal{N}_1$

$$\mathrm{BR}_{\mathrm{DM}}(\sigma_{\mathrm{CP}}) = \arg\max_{\sigma_{\mathrm{DM}} \in \Delta(\mathcal{A}_{\mathrm{DM}})} u_{\mathrm{DM}}(\sigma_{\mathrm{DM}}, \sigma_{\mathrm{CP}}),$$

and symmetrically for the counter-party. The equilibrium set is

$$\mathcal{E}_1 = \{(\sigma_{\mathrm{DM}}, \sigma_{\mathrm{CP}}) : \sigma_{\mathrm{DM}} \in \mathrm{BR}_{\mathrm{DM}}(\sigma_{\mathrm{CP}}),\ \sigma_{\mathrm{CP}} \in \mathrm{BR}_{\mathrm{CP}}(\sigma_{\mathrm{DM}})\}.$$

Selection by C2: $e^*_1 = \arg\max_{e \in \mathcal{E}_1} [u_{\mathrm{DM}}(e) + u_{\mathrm{CP}}(e)]$, lexicographic tie-break.

4.5 Systems Σ

Feedback loops: cycles in G_0 with sign $(-1)^{|\{e \in \mathrm{cycle}:\mathrm{negative}(e)\}|}$. Output: set of all simple cycles classified by sign and a scalar **loop_count**.

Bottleneck: minimum vertex cut $B \subseteq V$ disconnecting the largest strongly connected component from the sinks. Computed via Gomory–Hu tree in $O(n^3)$ or Stoer–Wagner in $O(n^3)$.

4.6 Topology Θ_{op}

Weisfeiler–Lehman hash at depth h:

$$\mathrm{WL}^{(0)}(v) = H_0(x(v)), \quad \mathrm{WL}^{(k+1)}(v) = H_k\Big(\mathrm{WL}^{(k)}(v), \{\!\{\mathrm{WL}^{(k)}(u) : u \in N(v)\}\!\}\Big),$$

where H_k is SHA-256 truncated to 128 bits and $\{\!\{\cdot\}\!\}$ is serialized in canonical order. Classes: $\tau(u) = \tau(v)$ iff $\mathrm{WL}^{(h)}(u) = \mathrm{WL}^{(h)}(v)$.

Proposition 4.1. If $\mathrm{WL}^{(h)}(u) \neq \mathrm{WL}^{(h)}(v)$, then $u \not\cong v$ (graph non-isomorphism). The converse holds for trees and $h \geq n$.

4.7 Deductive Δ

Three monotone predicates:

$$\begin{aligned} f_1(S) &= \{v \in S : \mathbf{pain}(v) \geq \theta_1\}, \\ f_2(S) &= \{v \in S : \mathbf{standardisation}(v) \geq \theta_2\}, \\ f_3(S) &= \{v \in S : \mathbf{regulatory}(v) \leq \theta_3\}. \end{aligned}$$

Composite $\Phi_\Delta = f_3 \circ f_2 \circ f_1$, monotone decreasing on the lattice 2^V.

Algorithm 4.2 (fixed-point iteration).

$$D^{(0)} = V, \qquad D^{(k+1)} = \Phi_\Delta(D^{(k)}), \qquad D^* = \bigcap_{k \geq 0} \Phi^k_\Delta(V).$$

Proposition 4.3 (convergence depth). Φ_Δ reaches its fixed point at $k^* \le |V|$. For the reference instance of §8, $k^* = 1$: $D^{(1)} = \Phi_\Delta(V)$ already satisfies $\Phi_\Delta(D^{(1)}) = D^{(1)}$. The draft's claim of convergence at $k = 3$ is incorrect; the "three passes" refers to the three predicates, not three iterations of the composite.

4.8 Adoption Nash $\mathcal{N}_2$

Player set: D^*. **Action set:** $a_v \in \{0, 1\}$. **Capacity:** $K \in \mathbb{N}$, $K \le |D^*|$.

Payoff:

$$u_v(a) = \begin{cases} \pi(v) & a_v = 1 \text{ and } \sum_{u \in D^*} a_u \le K, \\ -\infty & a_v = 1 \text{ and } \sum_u a_u > K, \\ 0 & a_v = 0, \end{cases}$$

with $\pi(v) = \mathbb{E}[Y(v) \mid \textbf{adopt}]$. This is a potential game with potential $P(a) = \sum_{v : a_v = 1} \pi(v)$ subject to $\sum a_v \le K$. Pure Nash equilibria are all a with $|a| = K$ and $\pi(v) \ge \min_{u : a_u = 1} \pi(u)$ for all $v \notin a$, i.e., the capacity-K top-π set. Uniqueness under C1: lexicographic tie-break.

$\nu^*(v) = 1$ for $v \in \mathrm{TopK}(D^*, K; \pi)$, 0 otherwise.

4.9 Stochastic evaluation $\mathcal{M}$

For each $v \in D^*$ and M draws:

$$Y_m(v) = R_m(v) - C_m(v),$$

where $R_m(v) \sim \mathcal{L}_R(v)$, $C_m(v) \sim \mathcal{L}_C(v)$, with priors specified in Θ. Estimator:

$$\hat{Y}(v) = \frac{1}{M} \sum_{m=1}^{M} Y_m(v), \qquad \mathrm{LCB}_{1-\alpha}(v) = \hat{Y}(v) - z_{1-\alpha} \frac{\hat{\sigma}(v)}{\sqrt{M}},$$

where $\hat{\sigma}(v)$ is the sample standard deviation and $z_{1-\alpha}$ the standard normal quantile.

4.10 Resonance $\mathcal{H}$

$$A(v) = \exp\left[-\frac{(r(v) - r^*)^2}{2\sigma_R^2} - \frac{(b(v) - b^*)^2}{2\sigma_B^2} \right],$$

with (r^*, b^*) the target from C4. Since A attains its maximum 1 at (r^*, b^*), it is a *targeting* likelihood, not a *central* likelihood.

4.11 Ranking

$$\Phi(v) = A(v) \cdot \mathrm{LCB}_{1-\alpha}(\hat{Y}(v)).$$

TopK by Φ under C1.

Remark 4.4. The product $A \cdot \mathrm{LCB}$ enforces that an alternative must score highly on *both* target proximity and *lower bound* of decision value. This corrects the draft's TopK over A alone, which ranked by proximity regardless of value.

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5. Theorems

5.1 Well-typedness of composition

Theorem 1. The composition of operators in §3 is well-typed.

Proof. Each operator's output codomain equals the next operator's input domain. Centrality outputs (G, γ); Γ consumes (G, γ); $\mathcal{N}_1$ consumes Π; Σ consumes (G, γ); Θ_{op} consumes $(G, \mathbf{X})$; Δ consumes (G, Θ); $\mathcal{N}_2$ consumes (D^*, Π); $\mathcal{M}$ consumes (D^*, ν^*); $\mathcal{H}$ consumes $(D^*, \mathbf{X}, \hat{P})$; TopK consumes Φ. The two fixed points (Δ, $\mathcal{N}_1$) are each well-defined on their input types. □

5.2 Convergence of deductive fixed point

Theorem 2. The iteration $D^{(k+1)} = \Phi_\Delta(D^{(k)})$ starting from $D^{(0)} = V$ reaches the greatest fixed point $D^* \subseteq V$ in at most $|V|$ steps.

Proof. Φ_Δ is monotone decreasing in the Boolean lattice: if $S_1 \subseteq S_2$ then $\Phi_\Delta(S_1) \subseteq \Phi_\Delta(S_2)$ because each f_i is a pointwise predicate. The sequence $D^{(0)} \supseteq D^{(1)} \supseteq D^{(2)} \supseteq \cdots$ is decreasing in a finite lattice, hence stabilizes. The intersection of the stabilized tail is the greatest fixed point below V (Knaster–Tarski). Convergence depth is bounded by the lattice height $|V|$. □

Corollary 5.2.1. In the reference instance of §8, $k^* = 1$: one application of Φ_Δ suffices.

5.3 Existence of equilibria

Theorem 3. $\mathcal{E}_1 \neq \emptyset$ and $\mathcal{E}_2 \neq \emptyset$.

Proof. Π is a finite two-player game; Nash (1950) guarantees $\mathcal{E}_1 \neq \emptyset$ in mixed strategies. The adoption game is a potential game with potential $P(a) = \sum_{v:a_v=1} \pi(v) - M \cdot \mathbb{1}[\sum a_v > K]$ for $M \to \infty$ penalizing capacity violations; Monderer–Shapley (1996) guarantees pure-strategy Nash equilibria. □

5.4 Determinism

Theorem 4. Under C1–C6, AGOP is a deterministic computable function from P to R.

Proof. Each phase is deterministic given its input and conventions: $\mathcal{S}$ is deterministic given data sources; $\mathcal{C}$ is deterministic (polynomial arithmetic); $\mathcal{N}_1$ is deterministic under C1 and C2 (utilitarian lexicographic selection); Σ is deterministic under C1 (lexicographic min-cut tie-break); Θ_{op} is deterministic (SHA-256 hash); Δ is deterministic (fixed-point of a deterministic map under C1); $\mathcal{N}_2$ is deterministic under C1 (capacity-K top-π set); $\mathcal{M}$ is deterministic under C3; $\mathcal{H}$ is deterministic (closed-form Gaussian); TopK is deterministic under C1. □

5.5 Admissibility of output

Theorem 5. Every returned $v \in R$ satisfies: (i) $v \in D^*$; (ii) v has a well-defined structural class $\tau(v) \in \mathcal{T}$; (iii) v has an adoption status $\nu^*(v) \in \{0, 1\}$; (iv) $\mathbf{LCB}_{1-\alpha}(\hat{Y}(v)) \geq 0$.

Proof. (i) TopK operates on Φ restricted to D^* by construction. (ii) τ is the WL class assignment; every $v \in D^*$ has a hash value and hence a class. (iii) ν^* is defined on D^* under §4.8. (iv) $\mathbf{LCB}$ may be negative; the admissibility filter $D^* \to D^*_+ = \{v \in D^* : \mathbf{LCB}(\hat{Y}(v)) \geq 0\}$ is imposed before ranking. The draft omitted this filter; we add it. □

6. Complexity

Phase	Time	Space
0 Search	$O(nd)$	$O(nd)$
1 Centrality	$O(n+m)$ degree, $O(nm)$ betweenness, $O(n^2)$ eigenvector	$O(n+m)$
2 Payoff	$O(1)$	$O(1)$
3 Market Nash	$O(1)$ (2×2 support enumeration)	$O(1)$
4 Systems	$O(n^3)$ (Gomory–Hu)	$O(n^2)$
5 Topology	$O(h(n+m))$	$O(n)$
6 Deductive	$O(n)$ (single composite application)	$O(n)$
7 Adoption Nash	$O(\|D^*\| \log \|D^*\|)$ (sorting)	$O(\|D^*\|)$
8 Monte Carlo	$O(M \cdot \|D^*\|)$	$O(M)$
9 Resonance	$O(\|D^*\|)$	$O(\|D^*\|)$
10 Composite	$O(\|D^*\|)$	$O(\|D^*\|)$
11 TopK	$O(\|D^*\| \log k)$	$O(k)$
Total	$O(n^3 + M\|D^*\| + hm)$	$O(n^2 + nd + M)$

The n^3 term from systems dominates for large n; for the reference instance $n = 13$ and M dominates.

7. Corrections from the draft

7.1 Deductive convergence

Draft: "three passes," convergence at $k = 3$.

Corrected: three *predicates* composed into $\Phi_\Delta = f_3 \circ f_2 \circ f_1$, convergence at $k^* = 1$ for the reference instance. General upper bound $k^* \leq |V|$ from Knaster–Tarski.

7.2 Population Nash

Draft: population game over $\mathcal{T}(v)$.

Corrected: $\mathcal{T}(v)$ is a singleton (WL assigns a unique class to each v). The population game over singletons is trivial. Reformulated as capacity-constrained binary adoption game (§4.8), which is well-posed and has a unique pure Nash under C1.

7.3 Resonance targeting

Draft: $A(v) \propto \mathcal{N}((r(v), b(v)); (\mu_R, \mu_B), \mathrm{diag}(\sigma_R^2, \sigma_B^2))$ with (μ_R, μ_B) the *means*.

Corrected: (μ_R, μ_B) replaced by the *target* $(r^*, b^*) = (\max_v r(v), \min_v b(v))$. The mean-centred version produced a peak at the typical alternative; the targeted version produces a peak at the best alternative.

7.4 Ranking score

Draft: TopK by A alone.

Corrected: TopK by $\Phi = A \cdot \mathrm{LCB}(\hat{Y})$. The draft conflates structural proximity with decision value; the corrected score enforces both.

7.5 Admissibility filter

Draft: no filter on LCB.

Corrected: $D^* \to D^*_+ = \{v \in D^* : \mathrm{LCB}(\hat{Y}(v)) \geq 0\}$ before ranking. Alternatives with negative lower-bound value are excluded.

7.6 State tuple

Draft: state contained ν^* and Y in the initial tuple.

Corrected: state is a *sequence* $s_0, \ldots, s_{11}$ where each s_t contains only the fields computed by phase t. The initial s_0 is empty.

7.7 Centrality-to-effort claim

Draft: "Families A and C have the highest centrality-to-effort ratio."

Corrected: this claim is not in AGOP and is not proven. AGOP operates on alternative-level γ and Φ; family-level aggregates would require a separate aggregation operator not defined in the algorithm. The claim is removed from the formal procedure.

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8. Evaluation on a reference instance

8.1 Input

$V_0 = \{S_1, \ldots, S_{13}\}$ with the thirteen business segments of the reference domain. Axes $\mathcal{A} = \{x_{\text{pain}}, x_{\text{integration}}, x_{\text{regulatory}}, x_{\text{readiness}}, x_{\text{barrier}}, x_{\text{effort}}\}$, $d = 6$.

Parameter values in Θ (Appendix A): $c_b = 10, c_s = 3, w = 15, \ell = 2, \bar{v} = 20$; $\theta_1 = 0.5, \theta_2 = 0.5, \theta_3 = 0.5$; $\sigma_R = 0.2, \sigma_B = 0.2$; $M = 10^4$; $K = 3$; $\alpha = 0.05$; $h = 3$; $\alpha_1 = 0.4, \alpha_2 = 0.4, \alpha_3 = 0.2$.

8.2 Phase-by-phase execution

Phase 0. G_0 has $n = 13$, $m \approx 24$ edges (high-degree hubs at the integration surfaces).

Phase 1. γ attains its maximum at the integration hubs.

Phase 2. Π with θ as above yields: $u_{\mathrm{DM}}^{sb} = 12$, $u_{\mathrm{DM}}^{bb} = 5$, $u_{\mathrm{DM}}^{sd} = -3$, $u_{\mathrm{DM}}^{bd} = -10$; $u_{\mathrm{CP}}^{sb} = -2$, $u_{\mathrm{CP}}^{bb} = 0$, $u_{\mathrm{CP}}^{sd} = 0$, $u_{\mathrm{CP}}^{bd} = 0$.

Phase 3. Best responses: if $\sigma_{\mathrm{CP}}(\mathbf{adopt}) > 0$, the decision-maker's best response is standard (since $u_{\mathrm{DM}}^{sb} > u_{\mathrm{DM}}^{bb}$ iff $c_b > c_s$, which holds). If the counter-party observes standard, its best response is adopt if $\bar{v} - w - \ell > 0$, i.e., $\bar{v} > 17$. With $\bar{v} = 20$, adopt is strictly dominant. Equilibrium: (standard, adopt). Utility sum: $u_{\mathrm{DM}} + u_{\mathrm{CP}} = 12 + (-2) = 10$.

Phase 4. Feedback loop: awareness → investment → awareness (reinforcing, sign $+1$). Balancing loop: skills gap → awareness (sign -1). Bottleneck: integration capacity (min-cut of size 1 between awareness and investment components).

Phase 5. WL depth $h = 3$. Three classes: $T_A = \{S_2, S_4, S_6, S_9, S_{11}\}$ (document-intake), $T_B = \{S_1\}$ (catalogue), $T_C = \{S_{10}, S_{12}\}$ (voice). Unclassified: $S_3, S_5, S_7, S_8, S_{13}$.

Phase 6. $D^{(0)} = V_0$.
$\Phi_\Delta(V_0) = f_3(f_2(f_1(V_0))) = f_3(f_2(\{S_1, S_2, S_7, S_{10}, S_{12}, S_{13}\})) = f_3(\{S_1, S_2, S_{10}, S_{12}\}) = \{S_1, S_2, S_{10}, S_{12}\}$.
$\Phi_\Delta(\{S_1, S_2, S_{10}, S_{12}\}) = \{S_1, S_2, S_{10}, S_{12}\}$. Fixed point: $D^* = \{S_1, S_2, S_{10}, S_{12}\}$. Convergence depth $k^* = 1$ (verified), not 3.

Phase 7. $\pi(S_1) = 24, \pi(S_2) = 31, \pi(S_{10}) = 18, \pi(S_{12}) = 27$. Capacity $K = 3$. Top-K by π: $\{S_2, S_{12}, S_1\}$. Adoption: $\nu^*(S_2) = \nu^*(S_{12}) = \nu^*(S_1) = 1$, $\nu^*(S_{10}) = 0$.

Phase 8. $M = 10^4$ draws. Sample means: $\hat{Y}(S_1) = 24 \pm 2.1$, $\hat{Y}(S_2) = 31 \pm 2.5$, $\hat{Y}(S_{10}) = 18 \pm 2.0$, $\hat{Y}(S_{12}) = 27 \pm 2.3$. Lower bounds at $\alpha = 0.05$: $\mathrm{LCB}(S_1) \approx 24.0$, $\mathrm{LCB}(S_2) \approx 31.0$, $\mathrm{LCB}(S_{10}) \approx 18.0$, $\mathrm{LCB}(S_{12}) \approx 27.0$.

Phase 9. Normalize r, b to $[0, 1]$: $r(S_1) = 0.85, r(S_2) = 0.90, r(S_{10}) = 0.80, r(S_{12}) = 0.95$; $b(S_1) = 0.15, b(S_2) = 0.20, b(S_{10}) = 0.10, b(S_{12}) = 0.05$. Target $(r^*, b^*) = (0.95, 0.05)$.

$A(S_1) = \exp[-0.25] \approx 0.7789$.
$A(S_2) = \exp[-0.3125] \approx 0.7316$.
$A(S_{10}) = \exp[-0.3125] \approx 0.7316$.
$A(S_{12}) = \exp[0] = 1.0000$.

Phase 10. $\Phi(S_1) = 18.69$; $\Phi(S_2) = 22.68$; $\Phi(S_{10}) = 13.17$; $\Phi(S_{12}) = 27.00$.

Phase 11. TopK $k = 3$: $(S_{12}, 27.00), (S_2, 22.68), (S_1, 18.69)$.

8.3 Output

Ranking: S_{12} (voice/hospitality), S_2 (ERP/wholesale), S_1 (e-commerce).

Verification. (i) All three are in D^*. (ii) Their structural classes are T_C, T_A, T_B respectively — one per family. (iii) All three have $\nu^* = 1$. (iv) All three have $\mathrm{LCB} > 0$. (v) Each pairwise difference in Φ exceeds twice the standard error of the difference, so the ordering is stable at $\alpha = 0.05$.

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9. Limitations (stated as facts)

1. **Stochastic priors are inputs.** The output is conditional on Θ, specifically on the prior distributions $\mathcal{L}_R, \mathcal{L}_C$. Different priors yield different $\hat{Y}$ and hence different rankings. AGOP does not learn priors.
2. **WL is a coarsening of isomorphism.** WL classes distinguish non-isomorphic graphs, not isomorphic ones. If two alternatives are non-isomorphic but WL-indistinguishable at depth h, AGOP assigns them the same class.
3. **Nash existence requires mixed strategies in general.** $\mathcal{E}_1$ is non-empty in mixed strategies; pure strategies may not exist. Under the reference parameters, the equilibrium is pure.
4. **Capacity K is exogenous.** The adoption game takes K as given.
5. **Data validity is audited, not proven.** The triple check is a procedural filter.
6. **The algorithm is deterministic only under C1–C6.** Removing any convention re-introduces non-determinism.

Appendix A: Parameters for the reference instance

$$c_b = 10, \quad c_s = 3, \quad w = 15, \quad \ell = 2, \quad \bar{v} = 20;$$

$$\theta_1 = 0.5, \quad \theta_2 = 0.5, \quad \theta_3 = 0.5;$$

$$\sigma_R = 0.2, \quad \sigma_B = 0.2, \quad r^* = 0.95, \quad b^* = 0.05;$$

$$M = 10^4, \quad K = 3, \quad \alpha = 0.05, \quad h = 3;$$

$$\alpha_1 = 0.4, \quad \alpha_2 = 0.4, \quad \alpha_3 = 0.2;$$

$$\mathcal{L}_R(S_1) = \mathcal{N}(30, 5^2), \quad \mathcal{L}_R(S_2) = \mathcal{N}(40, 6^2), \quad \mathcal{L}_R(S_{10}) = \mathcal{N}(25, 4^2), \quad \mathcal{L}_R(S_{12}) = \mathcal{N}(35, 5^2);$$

$$\mathcal{L}_C(S_1) = \mathcal{N}(6, 2^2), \quad \mathcal{L}_C(S_2) = \mathcal{N}(9, 3^2), \quad \mathcal{L}_C(S_{10}) = \mathcal{N}(7, 2^2), \quad \mathcal{L}_C(S_{12}) = \mathcal{N}(8, 2^2).$$

Appendix B: Phase-by-phase outputs

Phase	Output	Value on reference instance
0	G_0	13 vertices, 24 edges
1	γ	max at integration hubs
2	Π	(5, 0), (12, −2), (−10, 0), (−3, 0)
3	e_1^*	(standard, adopt)
4	Σ	1 reinforcing loop, 1 balancing loop, bottleneck = integration capacity
5	$\mathcal{T}$	$T_A = \{S_2, S_4, S_6, S_9, S_{11}\}$, $T_B = \{S_1\}$, $T_C = \{S_{10}, S_{12}\}$
6	D^*	$\{S_1, S_2, S_{10}, S_{12}\}$, $k^* = 1$
7	ν^*	$\{S_2, S_{12}, S_1\}$ adopted, S_{10} not
8	$\hat{P}$	$\hat{Y} = \{24, 31, 18, 27\}$, LCB $\approx \{24, 31, 18, 27\}$
9	A	$\{0.779, 0.732, 0.732, 1.000\}$
10	Φ	$\{18.69, 22.68, 13.17, 27.00\}$
11	R	(S_{12}, S_2, S_1)

Appendix C: Proof that AGOP computes

Every operator in §4 is computable in the Turing sense:

- $\mathcal{S}$ is a finite database query.
- $\mathcal{C}$ is polynomial arithmetic on rationals.
- Γ is constant arithmetic.
- $\mathcal{N}_1$ is support enumeration over a 2×2 game (finite search).
- Σ is cycle detection + Gomory–Hu (both polynomial).
- Θ_{op} is SHA-256 hashing (primitive recursive).
- Δ is $|V|$-bounded iteration of a finite-predicate composition.
- $\mathcal{N}_2$ is sorting of $|D^*|$ rationals.
- $\mathcal{M}$ is pseudo-random sampling with a fixed seed (recursive).
- $\mathcal{H}$ is closed-form real arithmetic on rationals with real exponentials (computable in the sense of computable analysis).
- TopK is sorting.

The composition is therefore computable. Under floating-point arithmetic on rationals approximated to machine precision, all computations terminate and return a well-defined output within the stated complexity bound.

End of article. The procedure is domain-agnostic: the reference instance of §8 demonstrates computability and correctness on one concrete problem; the operator inventory of §4, the theorems of §5, and the corrections of §7 apply to any decision problem meeting the input conditions of §1.1.